

B.Tech. Degree I Semester Regular Examination November 2023**SE 23-200-0103A ENGINEERING GRAPHICS**

(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Visualize and draw orthographic projection of straight lines and planes and solids.

CO2: Understand development of surface of different geometric shapes.

CO3: Construct isometric scale, isometric projections and views.

CO4: Obtain multiview projections and solid models of objects using CAD tools.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

- I. A straight line 65 mm long has one end 15 mm in front of VP. and 40 mm above H.P. while the other end is 30 mm in front of V.P. and 20 mm above H.P. Draw the plan and elevation of the line. What is the true inclination of the line with HP? What is the length of the plan? $(4 \times 12\frac{1}{2} = 50)$
- OR
- II. A circular lamina of 60 mm diameter is resting on H.P. on a point A which lies on its circumference. Draw the projections of the lamina when the lamina makes 45° with the H.P. and the diameter passing through A, is making an angle of 30° with V.P.
- OR
- III. A cone of base diameter 40 mm and axis length 60 mm has a point of its base circle in the VP, 40 mm above HP. Its axis is inclined at 45° to the VP and its front view is inclined at 35° to HP. Draw its projections.
- OR
- IV. A right circular cone of 100 mm base circle diameter and 50 mm height is resting on its base on the ground. It is cut by a vertical plane inclined at 40° to the V.P., the plane being at minimum distance of 15 mm, from the axis of the cone. Draw the sectional elevation. Also, project the true shape of the section. Assume that the section is in front of the axis.
- OR
- V. A square pyramid of base side 35 mm and axis 60 mm rests on its base on the ground with one of the sides of the base inclined at 30° to the VP. A string is wound round the surfaces of the pyramid starting from left extreme point on the base and ending at the same point. Find the shortest length of the string required. Also trace the path of the string in the front and top views.
- OR
- VI. A vertical cylinder of diameter 150 mm and 225 mm long is completely penetrated by a horizontal cylinder of diameter 126 mm and 210 mm long. The axis of the horizontal cylinder is parallel to VP., and is 10 mm in front of the axis of the vertical cylinder. Draw the projections of solids showing lines of intersection.
- OR
- VII. Draw the isometric projection of a sphere of diameter 16 mm kept centrally over a frustum of a square pyramid of height 25mm. The frustum has a base of side 35 mm and top of side 20 mm.
- OR
- VIII. A cylinder of diameter 40 mm and height 40 mm rests on the GP on one of its ends with its axis 35 mm behind the picture plane. The station point is 45 mm to the right of the axis. The station point is 65 mm above the GP and 40 mm in front of the PP. Draw the perspective view of the cylinder by visual ray method.

Bloom's Taxonomy Levels

L3 = 100%.